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(1) Research Program/Project/Study Title: Ethnoveterinary Medicine Among Ati Community in Municipality of Dumarao, Capiz, Philippines	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education/Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/Computer/Engineering/Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Dumarao
Province:		Capiz
Region:		6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Livestock		(8) Discipline/Field of Specialization Animal Science

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(9) Rationale

The use of EVM, or ethnoveterinary medicines, is one of the significant knowledge and practice among livestock and poultry farmers around the world. Ethnoveterinary medicine is a subset of traditional or indigenous knowledge of methods/practices concerned primarily with treating animal diseases and managing livestock. This form of medicinal treatment utilizes indigenous materials, such as herbs and other medicinal plants, in treating various animal diseases. Herbs possess therapeutic properties for both man and animals (Panda and Dhal, 2014).

For various ethnic groups, these EVMs play an important role in treating animal diseases. Herbs and other medicinal plants are accessible, and much cheaper than commercial treatments for they are located in the locality. Through the years, herbs and methods seen as effective were remembered and regularly used, while the ineffective ones were soon dropped and forgotten (Nadela, 2020).

As the economic development continues to rise in the country, ethnoveterinary medicine faces the risk of being forgotten due to the disappearance of traditional culture and natural resources. Thus, there is an urgent need to systematically document the traditional or indigenous methods being practiced as well as the medicinal plants being utilized (Ong and Kim, 2014). In the municipality of Dumarao, specifically in barangay Aglalana and Tamulalod, there is an indigenous group called Ati who still holds some of the traditional knowledge on ethnoveterinary medicine.

Hence, this study will be conducted to document the ethnoveterinary medicines used among Ati community in the municipality of Dumarao in order to preserve these knowledge as well as the methods being practiced in terms of traditional animal health care.

(9-A) Project Description

This study will be conducted to document the common ethnoveterinary plants and diseases of animals among Ati community in the municipality of Dumarao, Capiz. This study will gather information from the Ati respondents on the common diseases of their animals and the type of herbal plants they used to treat the disease. In addition, the proponents will also gather different herbal plants used by the Ati community for documentation and preservation. Prior to the start of the study, proper protocol in the conduct of the study to the community will be followed through coordination with the NCIP, Ati Tribe leaders and the municipality.

(10) Objectives

General Objectives

Evaluate the ethnoveterinary medicine among Ati community in Dumarao, Capiz, Philippines.

Specific Objectives

- Evaluate the socio-demographic and socio-economic profile of Ati community respondents.
- Document the different animal diseases and the ethnoveterinary medicine used by the Ati community.
- Collect and preserve different ethnoveterinary plants used by the Ati community.

(11) Review of Literature/Prior Art Search

One of the indigenous people groups in the country is the Negritos and they are considered as the earliest inhabitants of the Philippines. They are composed of about 25-34 tribal groups and are known in different names (Padilla, 2013). In Panay Island of Western Visayas the Negritos are called "Ati". The Ati communities can be found in the aggregated mountainous settlements of the island. They

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are located in six municipalities in Aklan, six municipalities in Antique (NCIP, 2019), 15 municipalities in Iloilo (PPDO, 2018), and one in Capiz.

The Ati from Panay is known as gatherers, peddlers, traders, and sellers of medicinal plants to the communities in the towns, cities, and nearby islands (de la Pena, 2009; Zayas, 2008).

Livestock and poultry raising indeed serve as a primary source of income for many indigenous people in the Philippines. Indigenous communities often engage in traditional agricultural practices, including raising livestock and poultry, as a means of subsistence and livelihood generation. These activities provide food security, income, and cultural significance within these communities. However, there are several diseases and factors that threaten the indigenous livestock and poultry raising that can cause substantial economic loss and food insecurity.

Ruminants, including cattle, goats, and sheep, are essential contributors to the production of meat, milk, wool, and other valuable products. However, they are susceptible to various diseases and other factors that can have detrimental effects on their health, welfare, and productivity. The following are some of the diseases and factors that pose threats to ruminants: lack of appetite, fever, cough and cold; diarrhea, dehydration, bloat, constipation and poisoning; intestinal parasites like tapeworm, whipworm, hookworm, roundworm and stomach worm; lice, ticks and mange; several infectious diseases such as anthrax, blackleg, and foot and mouth disease; and eye diseases, wounds, dog bites/rabies, burns, bleeding, snake bites, and difficulty in urinating, (IIRR, 1994).

On the other hand, among poultry, such as chicken, ducks, turkeys, and other domesticated birds, there are also several diseases and other factors which represent a diverse and complex challenge for indigenous poultry producers. These include: reduce appetite, diarrhea, intestinal worms, ticks, lice, mites, cough, and cold; fungal diseases like aspergillosis, aflatoxicosis, and trush; and various infectious diseases such as fowl cholera, fowlpox, infectious coryza, Newcastle disease, and salmonellosis (IIRR, 1994).

In addition, the Philippines has faced significant challenges in the swine industry due to the prevalence of various diseases among swine populations. These diseases not only pose a threat to the livelihood of farmers but also impact the country's food security and economy. Some of the most prevalent diseases and factors which greatly impact the health of swine are as follows: lack of appetite, fever, cough, colds, diarrhea, dehydration, constipation, and poisoning; internal parasites like lungworm, roundworm and tapeworm; lice, scabies or mite infestation; and other infectious diseases like foot and mouth disease, hog cholera, swine cholera, swine pox, etc. (IIRR, 1994).

Ethnoveterinary medicine (EVM) is a field for protecting animal health and treating illness that is associated with traditional beliefs and indigenous knowledge and practices (Khan et al., 2019).

Plant-based ethnoveterinary medicine is extensively used across the world since livestock raising is an integral part of most people's livelihoods. Approximately one percent of all known species (about 391,000) has a history of food use, while 10-15% are commonly used in traditional. Furthermore, ethnoveterinary medicine is very dynamic and multifunctional since it can cure a variety of livestock illness as well as being widely available in remote areas and less expensive than synthetic medicines (Willis, 2017).

Very little attention has been given to documentation of plants used as EVM, and this cultural reservoir is disappearing day by day in some parts of the developed world. However, in remote and

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underdeveloped areas in the world, the EVM is silently playing the vital role in sustainable livestock production (Khattak et al., 2015).

(12) Scientific Basis/Theoretical Framework

In many rural communities, especially among indigenous people, traditional knowledge of medicinal herbs has been passed down through generations, offering valuable resources for maintaining the health and well-being of livestock and poultry. Some of the medicinal herbs that are found locally in the rural communities that are used in treating diseases among livestock and poultry are as follows: prayer beads (*Abrus precatorius*), onion (*Allium cepa*), garlic (*Allium sativum*), aloe (*Aloe vera*), betel nut (*Areca catechu*), chili (*Capsicum annum*), papaya (*Carica papaya*), chrysanthemum (*Chrysanthemum indicum*), star apple (*Chrysophyllum cainito*), coconut (*Cocos nucifera*), lemon grass (*Cymbopogo citratus*), banaba (*Lagerstroemia speciosa*), sensitive plant (*Mimosa pudica*), bitter melon (*Momordica charantia*), rambutan (*Nephelium lappaceum*), rice (*Oryza sativa*), avocado (*Persea Americana*), guava (*Psidium guajava*), tamarind (*Tamarindus indica*), maize or corn (*Zea mays*), and ginger (*Zingiber officinale*) (IIRR, 1994).

(11) Methodology

Materials

The materials used in the study include the following: structured questionnaire, knife, cutting scissors, plastic bag/box for sample collection, camera and logbook.

Research Design. Purposive sampling will be used in the study in identifying the respondents. Households having animals (pets, ruminants, pigs, and poultry) will be chosen as the respondents of the study.

Location of the study. This study will be conducted in the two Ati communities in Barangay Tamulalod (Mt. Tag ao) and Aglalana, Dumarao, Capiz.

Survey Instrument. Interview schedule will be used to gather necessary data from the respondents. It will be written in English but will be translated in Hiligaynon during the survey.

Data Gathering Procedure. Prior to the start of the study, the proponents will follow the protocol in doing research activity in the IP community. Seeking the help of the NCIP, Municipal Tourism office, Barangay and the Ati community heads will be done. Respondents will be identified through the help of this people and respondents will be asked according to the structured questionnaire. The first part of the survey questionnaire will be the socio demographic and socio economic profile of the Ati respondents. The second part will be the different diseases of animals they encounter and the ethnoveterinary medicine they used. Documentation on the different species of these ethnoveterinary plants will also be conducted.

Common Animal Diseases and Ethnoveterinary Medicine in Ati Community

Data on the different animals encountered by the Ati respondents will be gathered. They will also be asked on what ethnoveterinary medicine they used to treat/control these diseases. Documentation on the picture and sample of these plants will be taken.

Preservation of the Ethnoveterinary Medicine used by Ati Community

The aim of this study is to preserve some of these herbal plants used by the community. The proponents will gather some of these plants for further identification of the plant experts and planting at the CapSU CVM for preservation.

Statistical Tools and Analysis

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Descriptive statistical method such as mean frequency, range and percentage will be used to analyze the data of the study.

(14) Major Activities/Work plan

RESEARCH ACTIVITIES	Jan	Feb	Mar	Apr	May	Jun	Jul-Dec
Acquisition of materials							
Data Gathering and Analysis							
Technical reports/presentations							
Preparation of a Ethnoveterinary Medicine Booklet							
Copyright Application of the Booklet prepared							
IEC materials/Flyers production							

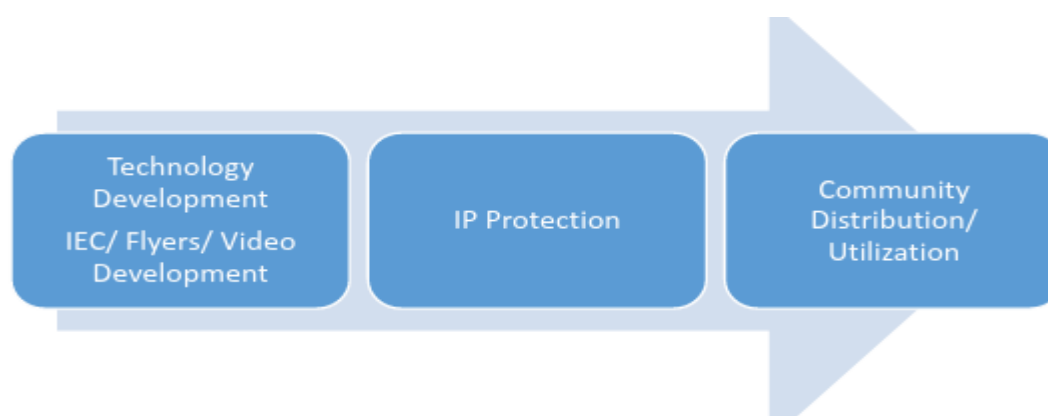
(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Generate job through hiring of research assistant Help tourism people for the documentation of the ethnoveterinary medicine among Ati community	1
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials Develop Ethnoveterinary Medicine Booklet	2
Patents/IP	File 2 copyrights for the developed IEC materials and videos.	2
Places and Partnerships	Forge linkages and partnerships to LGU's, NCIP, Capiz Tourism Office, Dumarao LGU, and other organizations.	5
Products	Ethnoveterinary Medicine Booklet Paper for Publication Preserve ethnoveterinary medicine	
Policy	Develop one policy inputs for the preservation of the ethnoveterinary medicine among Ati community	1
Impact to Social/ Economics	<i>Social Impact:</i>	

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	Information and awareness of the ethnoveterinary medicine among Ati community <i>Economic Impact:</i>		
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(15) Technology Transfer Activity Design



(16) Targets Beneficiaries

- Ati Community
- NCIP
- Department of Tourism

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Study Leader	Animal Science
Dr. Marie Jean P. Penson	Member	
Joshua B. Gaspar	Member	

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (CapSU)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 @ 11,068 x 6 months	66,408.00
B. Maintenance and Other Operating Expenses (MOOE)		
Transportation allowance		30,000

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Representation Expenses		30,000
Training Expenses		40,000
Supplies (Office supplies and Foods/ Give aways to Ati Respondents)		50,000
Total		216, 408.00

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